Mini Project 1 Report

Exploratory Data Analysis

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Chapter 1

Introduction

Exploratory Data Analysis (EDA) is an essential step in the data science workflow used to understand datasets before applying advanced analytics or machine learning techniques.

This project analyzes a Customer Analytics Dataset containing customer demographic details, income information, purchasing behavior, and online activity patterns.

Each row in the dataset represents an individual customer, while columns describe different characteristics of that customer.

The objectives of this project are:

- Understand dataset structure
- Perform data cleaning and preprocessing
- Explore data distributions
- Identify relationships between variables
- Generate meaningful insights

Chapter 2

Dataset Description

The dataset contains 250 customer records and 14 attributes after preprocessing.

2.1 Important Features

- CustomerID – Unique identifier
- Age – Customer age
- Gender – Male/Female
- City – Customer location
- Education – Education level
- MaritalStatus – Marital status
- AnnualIncome – Yearly income
- SpendingScore – Spending behavior score
- YearsEmployed – Work experience
- PurchaseFrequency – Number of purchases
- OnlineVisitsPerMonth – Website visits
- ReturnedItems – Number of returned products
- PreferredDevice – Shopping device
- LastPurchaseAmount – Last transaction value

Chapter 3

Methodology

3.1 Phase 1: Setup and Inspection

The dataset was loaded using Python Pandas. Initial inspection was performed using:

- head()
- info()
- describe()

These methods helped understand data types, structure, and statistical properties.

3.2 Phase 2: Data Preprocessing

Data cleaning steps included:

- Identification of missing values using isnull()
- Education column filled using mode value
- AnnualIncome column filled using median value
- Duplicate rows removed

These steps ensured data reliability and consistency.

3.3 Phase 3: Exploratory Data Analysis

3.3.1 Univariate Analysis

Univariate visualizations included:

- Age distribution histogram
- Gender distribution bar chart
- Annual income histogram

3.3.2 Bivariate Analysis

Relationships between variables were analyzed using:

- Scatter plot of Age vs Spending Score
- Boxplot of Gender vs Spending Score

3.4 Phase 4: Multivariate Analysis

A correlation matrix was generated and visualized using a Seaborn heatmap to analyze relationships among multiple numerical variables.

Chapter 4

Results and Discussion

- Majority of customers belong to the 30–45 age group.
- Spending score shows weak dependency on annual income.
- Higher online visits are associated with increased purchase frequency.
- Purchasing behavior is balanced across genders.

Chapter 5

Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Chapter 6

Challenges Faced

- Handling missing values appropriately
- Selecting correct visualization techniques
- Interpreting correlations carefully

Chapter 7

Conclusion

The project successfully performed exploratory data analysis on the customer analytics dataset. Data preprocessing improved dataset quality, while visualizations helped uncover meaningful customer behavior patterns. The cleaned dataset can be further used for machine learning and customer segmentation tasks.

Chapter 8

Future Scope

- Customer segmentation using clustering
- Predictive analytics using machine learning
- Personalized marketing recommendation systems

Chapter 9

References

- Pandas Documentation
- Seaborn Documentation
- Python Data Analysis Tutorials